

A/B Test: Subscription Screen with 50% Discount Label

Analysis and Test Planning

Context

Current Screen Conversion	17% of users who reach the subscription screen
Subscription Screen Reach	34% of all new installs
Daily Installs	~2 000 users per day
Subscription Price	\$4.99 / week
Tested Change	Adding a "50% Discount" label on the subscription screen (actual price remains \$4.99)

Hypotheses

Null Hypothesis (H_0)

Adding "50% discount" information to the subscription screen will not affect the purchase conversion rate; any observed difference between the variants will be statistically insignificant

Alternative Hypothesis (H_1)

Users who see the "50% discount" offer (at the same \$4.99 price point) will purchase the subscription more frequently due to the psychological framing of the deal, leading to an increase in the install-to-paid conversion rate

Test Audience

Inclusion Criteria

New users installing the app for the first time

Exclusion Criteria

Users with an active or recently cancelled subscription

Randomization

Users are randomly split into groups (50/50) at the moment of app installation

Metrics

Target Metric (Primary)

Conversion Rate (Full Funnel, Install → Purchase): The percentage of all new users (from the moment of installation) who completed a subscription purchase

Target Metric (Primary)

Retention: Monitoring if the quality of acquired users has decreased

Renewal Rate: Tracking whether users continue their subscription after the first week.

Screen Conversion: The percentage of purchases specifically among those who reached the discount screen

Subscription Cancellation Rate (7d): Percentage of buyers who cancelled within 7 days. The discount might attract less motivated users who churn quickly, potentially lowering long-term revenue

Sample Size and Test Duration

Parameters

Baseline Conversion

$34\% \text{ (reach)} \times 17\% \text{ (purchase)} = 5.78\%$

Minimum Detectable Effect (MDE)

20% (relative uplift)

Expected New Conversion

$5.78\% \times 1.2 = 6.936\%$

Significance Level (α)

5%

Statistical Power ($1-\beta$):

80%

Required Sample Size

Based on calculation, **5 508 users** per group.

Test Duration

11 016 total users / 2 000 daily installs $\approx$ 5.5 days. The test will run for **7 days** to cover a full weekly cycle and account for seasonality

Success Criteria and Actions

✓ Successful Test

The result is statistically significant (p-value < 0.05).

The primary metric (Conversion Rate) is higher in the experimental group without a significant drop in secondary metrics.

Action:

Roll out the new "discount label" design to 100% of users.

✗ Unsuccessful Test

No statistically significant difference is found, or the result is negative

Action:

Retain the current version. Analyze why the hypothesis failed and test an alternative

Alternative Ideas

Free Trial

Implement a 3-day free trial to allow users to experience premium features before the first charge

Annual Subscription

Test a yearly subscription plan as an alternative to the weekly model

Social Proof

Add "active user counts" or a "customer reviews" block to the payment screen to build trust

Limited-Time Offer

Implement a countdown timer for the special offer to stimulate faster decision-making



AB Testguide

How many visitors do you need?

Conversion rate Control
via your test page, in %

Expected improvement over control
relative, in %

Unique visitors on your test page per week

Max number of weeks for AB-test
used to calculate minimum expected relative
improvement vs Control

Hypothesis:

- ☒ One-sided
☐ Two-sided

Power:

- ☐ 75%
☒ 80%
☐ 90%
☐ 95%

Required confidence level (1 - alpha):

- ☐ 90%
☒ 95%
☐ 99%

Minimum sample size:

5508

unique visitors per test variation

Power

Confidence level

CR A

CR B

AB test duration

Minimum test duration 0.79 weeks *

Round up to a **AB-test period of 1 weeks** (discrete number of business cycles)

Minimum improvement needed

Minimum Conversion rate B of 6.8 % needed to run AB-test in max 1 weeks, with 14000 unique visitors per week on your page *

This amounts to a **minimum relative improvement of 18% needed** in order to run the test at the required levels of power and confidence.

* : assuming an AB-test with two variants (Control and B)

Both calculations use R-function *power.prop.test* as described in

[37signals.com blog AB-testing tech note determining sample size](http://37signals.com/blog/AB-testing-tech-note-determining-sample-size/)

[<< Back to the power and confidence visualisation tool](#)

Recommended reading

<http://www.win-vector.com/blog/2014/05/a-clear-picture-of-power-and-significance-in-ab-tests/>

Data Input

Confidence level %:

Power Level %:

Conversion Rate for control %:

Minimum Detectable Effect (MDE) %:

Number of Variants (not including control):

Weekly Traffic:

Is the Minimum Effect Relative?

☒ Yes
☐ No

One-sided or Two-sided Test?

☒ One-sided
☐ Two-sided

Calculate Sample Size

Results

Sample Size per Group:	5195
Total Sample Size:	10390
Estimated Duration (weeks):	1